Sprague-Grundy theorem

Let's consider a state v of a two-player impartial game and let v_i be the states reachable from it (where $i \in \{1, 2, \dots, k\}, k \ge 0$). To this state, we can assign a fully equivalent game of Nim with one pile of size x. The number x is called the Grundy value or nim-value of state v.

Moreover, this number can be found in the following recursive way:

$$x = \operatorname{mex} \{x_1, \ldots, x_k\},$$

where x_i is the Grundy value for state v_i and the function mex (*minimum excludant*) is the smallest non-negative integer not found in the given set.

Viewing the game as a graph, we can gradually calculate the Grundy values starting from vertices without outgoing edges. Grundy value being equal to zero means a state is losing.